

AHMED FERGANEY

Quality Assurance & Process Engineer

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PROFILE

Quality-focused Process Engineer with hands-on experience in supplier and incoming quality assurance, inspection, and defect resolution across the automotive and home appliance manufacturing sectors. Proven record of reducing defect rates through structured root-cause and corrective-action methods (8D, RCA, CAPA), supervising quality/inspection teams, and driving continuous improvement to meet company, customer, and regulatory quality standards.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department
- Graduation Project: Design, Modeling, and Control of a Six-DOF Artificial Upper Limb

PROFESSIONAL EXPERIENCE

Product Development Engineer | Kandil Glass | 10th of Ramadan City, Egypt06/2025 – Present

- Led end-to-end development of customized glass products, from initial client requirements (via sales) through feasibility analysis, design, and full-scale manufacturing.
- Managed cross-functional execution across design, production, and operations; monitored production performance and analyzed quality data to optimize efficiency and delivery.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt02/2025 – 05/2026

- Led a work standardization (Standard Work) project across production lines, improving Line of Balance (LOB) from 60% to 75%, resulting in enhanced line efficiency and better workload distribution — within the Home Appliances sector.
- Directly supervised and coached a production/process team, conducting training and performance reviews to build capability and sustain quality standards.
- Led cross-functional training and process transformation initiatives, enabling knowledge scaling and collaboration across teams.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt12/2023 – 12/2024

- Reduced defect rate from 8% to 2% (75% reduction) on the VW automotive assembly line by leading root cause analysis (RCA), PFMEA, 8D reporting, and corrective/preventive actions (CAPA).
- Directly supervised a quality inspection team, coordinating daily inspection activities and verifying non-conforming material was contained before reaching production.
- Recorded and managed inspection results and quality documentation using the site's ERP/QMS system, ensuring traceability and audit-readiness.
- Coordinated with internal suppliers on defect containment, corrective and preventive actions (CAPA), and proactive prevention plans; internal supplier relayed requirements to external suppliers to close the loop on recurring quality issues.
- Applied structured problem-solving methodologies (DMAIC, DFMEA, 5 Why, Fishbone) to diagnose recurring failures and verify effectiveness of corrective actions (RCCA).

Continuous Improvement Engineer | JAC | Badr City, Egypt03/2023 – 11/2023

- Responsible for quality and continuous improvement on the Water Dispenser and TV production lines, within the Electrical Appliances sector.
- Increased Water Dispenser line productivity from 150 to 225 units per 2-hour cycle (50% improvement) through process and workflow optimization.

CORE QUALIFICATIONS

Quality & Standards: Supplier Quality Coordination, 8D, Root Cause Analysis (RCA), RCCA, 5 Why, Fishbone/Ishikawa, CAPA & Proactive Prevention Plans, PFMEA/DFMEA & Control Plans, Sampling Plans & Quality Control Procedures, Pareto Analysis, Audit Participation & Regulatory Compliance

ERP / Documentation Systems: Inspection recording & quality documentation via ERP/QMS platforms; proficient in Microsoft Office (Excel, PowerPoint, Word)

Process Engineering & Manufacturing: Process design, mapping & optimization (VSM, SIPOC, process flow diagrams), line balancing, capacity planning & takt-time analysis, work standardization & SOP development, time & motion studies, cycle time reduction

Continuous Improvement & Lean: Lean Manufacturing (Kaizen, 5S, SMED, Waste Reduction, DM), Six Sigma fundamentals (DMAIC), KPI management (OEE, FPY, scrap rate, downtime), cost reduction & productivity initiatives

Data & Analysis: Python for data analysis, SQL for production/quality data querying, Statistical Process Control (SPC, P-Value, control charts, hypothesis testing), data visualization

Leadership: Supervised and coached quality/inspection technicians, cross-functional team leadership, training & performance reviews, technical mentoring, problem-solving under constraints

CERTIFICATIONS

- Oxyacetylene and ARC-Welding in High-Pressure Pipes Diploma | Don Bosco Institute | 2019
- Turning Course | Don Bosco Institute | 2020
- Automation Basic Diploma | HA Consulting Group | 2020
- Robotics Camp with a Case Study (Robotic Arm) | RTE | 2020
- Mechanics of Automotive Diploma | Don Bosco Institute | 2021
- Lean Six Sigma Yellow and Green Belt | Udemy | 2022
- CPIM Part One | HPA | 2023
- Embedded System Diploma | Eng. Ahmed Abd El-Ghafar | 2023
- Statistics, Linear Algebra & Numerical Optimization | ITI | 2024
- Databases, Power BI, Python for Data Analysis | Udemy | 2025
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, AI Agents | Udemy | 2026

LANGUAGES & TOOLS

Languages: Arabic (Native), English (Professional), German (Basic)

Tools: AutoCAD, Autodesk Inventor, ERP/QMS systems, Power BI, MS Office, Jira, Trello, MATLAB